

A Sample a task



Native TabICL SCM prior

generates features X and continuous score(s)



One synthetic episode with multiple regimes

Regime colors are generator labels; they are not model inputs.

x_1	x_2	Y	split
-0.8	0.2	0	support
-0.4	0.7	1	support
0.1	-0.6	0	support
0.5	0.3	1	support
0.8	-0.1	—	query
1.1	0.6	—	query

Support Y is given; query Y is hidden.

C Route rows to regimes

Either routing method can pair with either mechanism.

soft gate

Observed x_i sets the regime.

Sort support x_i ; split its ranks into bins.
Query x_i uses those same boundaries.



Support $x_i \rightarrow$ assign Z

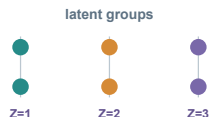
Query $x_i \rightarrow$ same Z cuts

Z is determined by observed X_i .

persistent

Group identity sets the regime.

Rows in a group keep one assignment.
The group ID is not included in X .



same group $\rightarrow$ same regime

Group ID is not included in X .

Membership stays latent.

B Score and label mechanisms

Example numbers are schematic; class IDs may be permuted.

r_z

SCM gives each regime its own score and cut-point draw

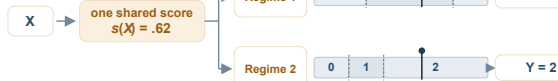
Regime 1 $X \rightarrow s_1(X) = .62 \rightarrow 0 \quad 1 \quad 2 \rightarrow Y = 1$

Regime 2 $X \rightarrow s_2(X) = .30 \rightarrow 0 \quad 1 \quad 2 \rightarrow Y = 1$

Rule settings match; sampled cut points are not shared.

g_z

One shared score; each regime uses its own cut points



Same score (.62); different cuts give different labels.

D Pretrain and predict

Same in-context prediction task for all schedules

Native control

single-regime tasks

Fixed mixture

30% multiregime

Curriculum

share ramps 0 $\rightarrow$ 50%

Prediction uses labelled examples plus the query features

Model input

support: X_s, Y_s

query: X_q only

NanoTabPFN

in-context predictor

Query prediction

$p(Y_q | X_s, Y_s, X_q)$

query labels are not supplied

Primary prediction input omits the regime ID.

Soft-gate membership may be inferred from X_i ; persistent group membership remains latent.